VIJAY VARSHAN RAJ G J

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SUMMARY

Final-year Computer Science undergraduate (B.Tech, graduating 2027) and AI/ML engineer who implements, integrates, and deploys machine learning models as production-ready, API-driven applications. Hands-on with Python, TensorFlow, scikit-learn, Pandas, NumPy, SQL, Git, and REST APIs, with three end-to-end deep learning projects (CNN, transfer learning, LSTM) delivered as deployed web apps with measured accuracy and inference latency. During an ML / Software Engineering internship, built a Python/Flask backend with 8+ REST API endpoints, root-caused a MySQL/MongoDB integration failure, and improved reliability through input validation and unit testing in a 5-member Agile team using Git-based code review. Strong foundation in data structures, algorithms, OOP, and DBMS, with a focus on explainable and responsible AI (Grad-CAM, LIME). Backed by an IEEE publication and a second paper under review.

EDUCATION

B.Tech in Computer Science and Engineering — Sep 2023 – May 2027

SRM Institute of Science and Technology, Tiruchirappalli, Tamil Nadu | CGPA: 8.45

TECHNICAL SKILLS

AI/ML & Data: TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, CNN, LSTM, transfer learning, data preprocessing, model evaluation & optimization, Grad-CAM, LIME, OpenAI API

Programming & CS Fundamentals: Python, C, JavaScript, PHP, SQL, data structures & algorithms, OOP, DBMS, software engineering fundamentals

Model Integration & Deployment: Flask, REST API design & integration, JWT session management, AWS, Git, GitHub, unit testing, Agile development, production deployment

Databases: MySQL, MongoDB, Firebase (SQL & NoSQL design, query optimization, schema modeling)

Agentic & AI-Assisted Development: Claude Code, Cursor, GitHub Copilot (daily agentic coding workflows for feature development, debugging, and refactoring)

Frontend & Mobile: React.js, Node.js, HTML5, CSS3, Bootstrap, Flutter, React Native

EXPERIENCE

ML / Software Engineering Intern · Inforios Software Technologies, Tanjore, India — Dec 2025 – Jan 2026

Stack: Python, MySQL, MongoDB, Git, REST APIs

- Implemented the Python/Flask backend for a livestock production tracking and inventory management system end-to-end, tracing logs to root-cause a MongoDB connectivity failure and restoring reliable data flow before delivery.
- Designed, built, and documented 8+ RESTful API endpoints integrating frontend modules with MySQL/MongoDB data services, cutting integration bugs by 20% through clearer contract definitions.
- Improved reliability by cutting recurring production errors by 25% through input validation middleware and unit tests, working in a 5-member Agile team with Git-based code review.

PROJECTS

XAI X-Ray Analysis — Full-Stack Medical Image Classification Web App — Feb 2026 – May 2026

Python | Flask | TensorFlow | CNN | MySQL | Bootstrap | SQLAlchemy — deployed, multi-user web app

- Built and deployed a TensorFlow CNN chest X-ray classifier as a multi-user Flask REST application (SQLAlchemy/MySQL, Bootstrap, JWT sessions), achieving 98% accuracy on a 5,216-image dataset with sub-3-second inference for 10+ concurrent users.
- Integrated Grad-CAM and LIME explainability so users could validate model predictions in under 30 seconds, reducing review overhead by 45%.

FreshLens AI — Meat Quality Assessment System — Sep 2025 – Dec 2025

Python | TensorFlow | MobileNetV2 | OpenCV | Streamlit | OpenAI API

- Fine-tuned a MobileNetV2 CNN via transfer learning to reach >95% meat freshness classification accuracy at sub-5-second inference on a custom 4,000+ image dataset.
- Architected a modular IoT sensor-fusion interface with standardized data contracts and decoupled microservice design, cutting estimated future model-integration time by 60%.

Hybrid Intrusion Detection System (IDS) — May 2025 – Jul 2025

Python | TensorFlow | Streamlit | LSTM | Scikit-Learn | Finite Automata

- Engineered a hybrid IDS combining LSTM deep learning with deterministic finite automata rule engines, achieving 99.2% binary and 97.8% multi-class accuracy on the CICIDS2017 benchmark.
- Scaled the pipeline to classify 2.8M+ network traffic records in real time with sub-200ms average detection latency, and deployed a Streamlit monitoring dashboard that cut mean time-to-detect by 35%.

CERTIFICATIONS

- Machine Learning, Data Science & Generative AI with Python — Coursera
- Generative AI & Machine Learning — AWS
- Virtual Internship: AI/ML — EduSkills (AICTE)
- MERN Full Stack Development — Coursera

RESEARCH & PUBLICATIONS

- **IEEE Published:** Transparent Secure Supply Chain Packing — ICCMC 2025 · IEEE Xplore
- **Under Review:** Carnisight 2.0 — Next-Gen Freshness Estimation of Perishable Foods — ICCIT 2025